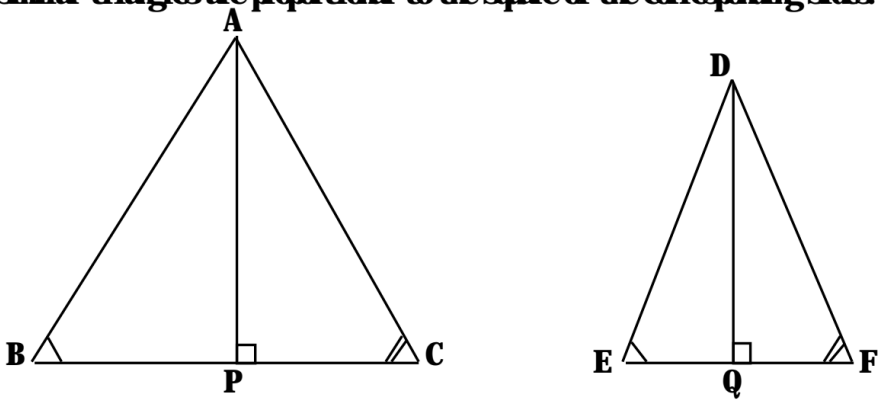
10. Area (Similar Triangles)

Theorem

The areas of similar triangles are proportional to the square of the corresponding sides.



Given : **Triangles ABC, DEF are similar.**

To prove : $\dfrac{\text{Area of triangle ABC}}{\text{Area of triangle DEF}} = \dfrac{BC^2}{EF^2}$

Construction : **Draw AP perpendicular to BC and DQ perpendicular to EF.**

Proof : **In triangles APB, DQE**

$\angle ABP = \angle DEQ$ (given)

$\angle APB = \angle DQE$ $(= 90^0)$

Therefore third angle in each are equal.

Triangles APB, DQE are similar.

$\Delta^s APB, DQE$ ///

$\dfrac{AP}{DQ} = \dfrac{AB}{DE}$ **But** $\dfrac{AB}{DE} = \dfrac{BC}{EF}$

Therefore $\dfrac{AP}{DQ} = \dfrac{BC}{EF}$

Now, $\dfrac{\text{Area of triangle ABC}}{\text{Area of triangle DEF}} = \dfrac{\frac{1}{2}BC \times AP}{\frac{1}{2}EF \times DQ}$

$= \dfrac{BC}{EF} \times \dfrac{AP}{DQ}$

$= \dfrac{BC}{EF} \times \dfrac{BC}{EF}$

$= \dfrac{BC^2}{EF^2}$

Excercise 10

1 **ABC is a right angled triangle, right angled at A. AD is perpendicular to BC.**

Prove that $\Delta BAD : \Delta ACD = BA^2 : AC^2$

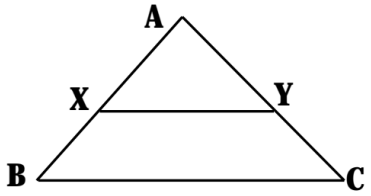
2 **In trapezium ABCD, AB is parallel to DC. AC and BD interesect at O. If AO** $= \frac{1}{4}$ **AC, prove that** $\Delta AOB = \frac{1}{9}\Delta COD$

3 **ABC is an isoceles triangle right angled at A. Outside the triangle ABC, ABD and BCE are two equalateral triangles on AB and BC respectively.**

Prove that $\Delta ABD : \Delta BCE = 1:2$

4 **ABCD is a trapezium. AB is parallel to CD. If diagonals intersect at O and AB = 2cm find the ratio between** Δ **AOB and** Δ **COD.**

5 **ABC is a triangle. XY is parallel to BC. If** $\Delta AXY : fig\ XBCY = 4:5$ **show that** $AX : XB = 2:1$.



6 **ABC is an acute angled triangle. BD, CE are altitudes. BD, CE intersect at X. Fill in the blanks**

a) $\Delta ABD : \Delta ACE = BD^2 :$ .....

b) $\Delta BXE : \Delta CXD =$ ..... $: CD^2$

c) $\Delta ABD :$ ..... $= AD^2 : AE^2$

d) $\Delta BXE :$ ..... $= BX^2 : CX^2$